STAFF REVIEW · H446 001 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Do Now — source answers

1. Integer for 27; real / floating point for 18.5. A count uses whole numbers. A temperature may contain a fractional part.

2. String . The value is an identifier rather than a number for calculation. A string preserves the leading zeros; an integer would store the numerical value 42.

3. 45 32 + 8 + 4 + 1 = 45

4. There are 256 different bit patterns , but the unsigned range starts at 0. Eight bits represent 0–255 . The value 256 needs 9 bits: 100000000_2 .

I Do — source worked steps

1. Write the columns — CODE:
128 64 32 16 8 4 2 1
0 0 1 0 1 1 0 1

2. Use only the columns containing 1 — CODE:
32 + 8 + 4 + 1

3. Calculate the denary value — CODE:
32 + 8 + 4 + 1 = 45

We Do — guided CHECK

1. Build the justification

2. Choose the place values
74 = 64 + 8 + 2.
Check: 64 + 8 + 2 = 74.

3. Build the answer

4. Complete the rule

You Do — source mark schemes

1. CODE:
number of readings → Integer
Reason: a whole-number count.

air temperature, e.g. 18.5 → Real / floating point
Reason: may contain a fractional part.

one-letter category → Character
Reason: one character is stored.

station identifier "0042" → String

Reason: it is an identifier and the leading zeros must be preserved.

sensor working → Boolean
Reason: only two states are required: true/false.

2. CODE:
37 = 32 + 4 + 1
37 = 00100101

92 = 64 + 16 + 8 + 4
92 = 01011100

200 = 128 + 64 + 8
200 = 11001000

3. CODE:
01101001
= 64 + 32 + 8 + 1
= 105

11000010
= 128 + 64 + 2
= 194

4. CODE:
Largest unsigned 8-bit value:
2^8 - 1 = 255

8 bits are insufficient because:
300 > 255

9 bits give:
0 to 2^9 - 1
= 0 to 511

Minimum required width = 9 bits

Exit Ticket — source answer

1. True. Eight bits give 256 values, from 0 to 255.